

SciFig-Evaluation: A Multilingual Benchmark and Behavioural Framework for Evaluating Multimodal Language Models on Scientific Figures

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Declaration

No portion of the work contained in this document has been submitted in support of an application for a degree or qualification of this or any other university or other institution of learning. All verbatim extracts have been distinguished by quotation marks, and all sources of information have been specifically acknowledged.

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Date: 2026

Abstract

Multimodal large language models are increasingly asked to read scientific figures, yet existing benchmarks largely measure whether a model can extract a value from a chart, not whether it is a faithful visual reader. This thesis introduces SCIFIG-EVAL, a multilingual benchmark of 1,005 figures drawn from 158 arXiv papers and paired with 1,411 expert annotations in English, Bulgarian, German and Chinese, together with a three-axis behavioural framework — Admittance, Resistance and Inductance — that separates honesty in the absence of evidence, faithfulness under misleading context, and sound inductive reasoning. Using an MQM-style multi-judge evaluation pipeline, the framework is exercised on eleven contemporary models spanning five families via five adversarial probe types. The resulting behavioural profile reveals systematic, near-orthogonal failure modes that conventional accuracy scores obscure.

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Contents

1	Introduction	6
1.1	<i>Homo sapiens</i> : A Species of Visual Thinkers	6
1.2	Can <i>Intelligentia Machinae</i> (Machine Intelligence) Comprehend Visuals Similarly?	8
1.3	Motivation for Focusing on Scientific Research Figures	8
1.4	Research Questions	9
1.5	Thesis Structure	10
2	Background	11
2.1	How do today’s Large Language Models see?	11
2.2	The scientific figure as visual argument	12
2.3	Failure modes of visual reading	13
2.4	Open-ended evaluation and the LLM-as-judge	14
2.5	Reading across languages	14

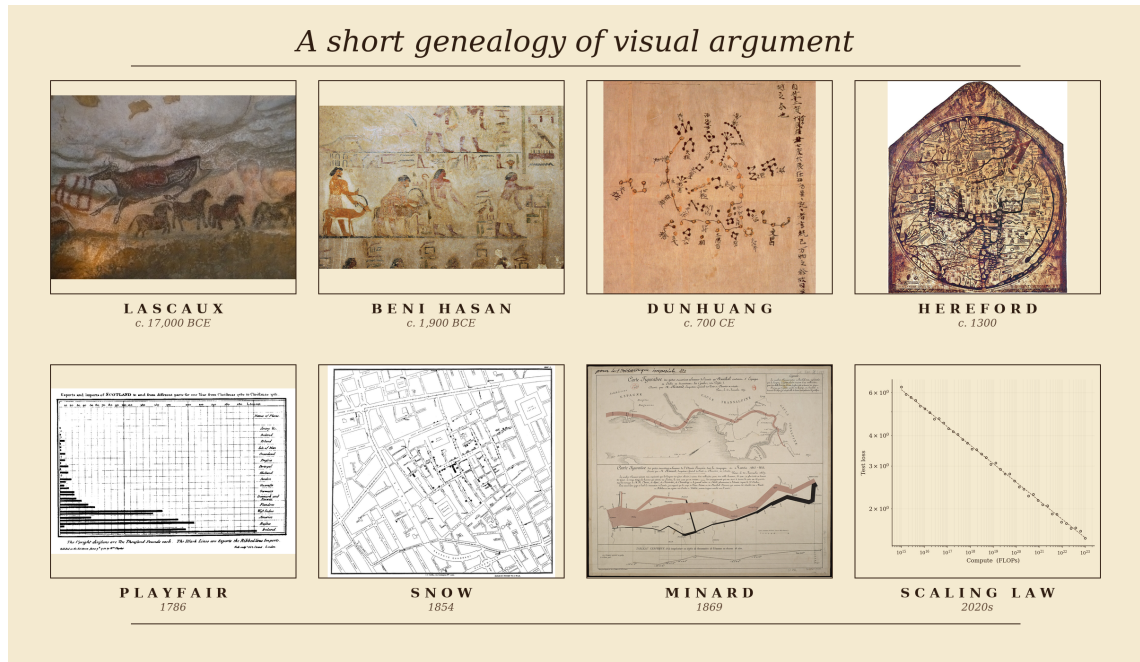
Chapter 1

Introduction

1.1 *Homo sapiens*: A Species of Visual Thinkers

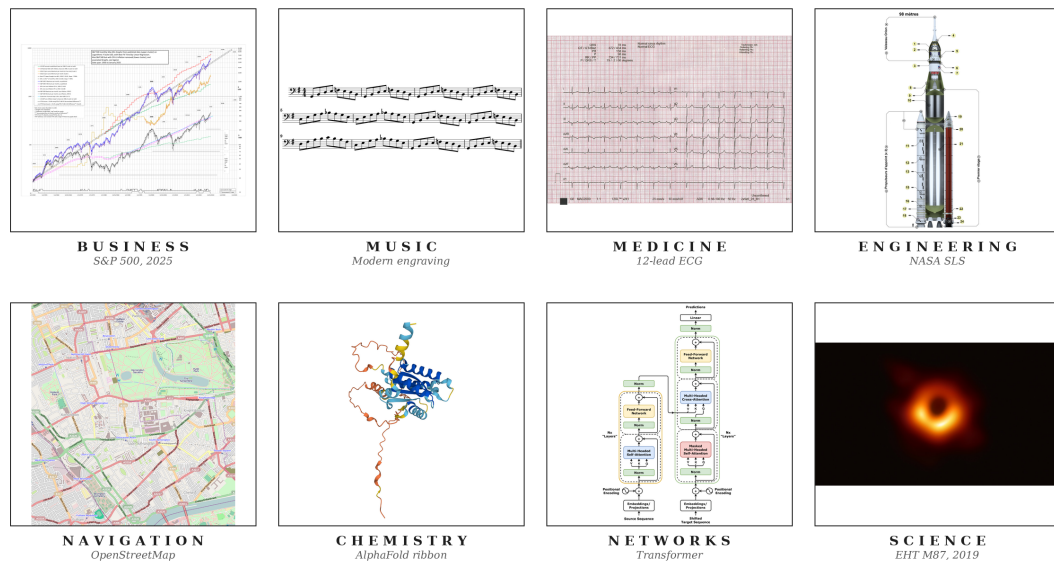
The record of a species willing to reason pictorially begins no later than the Upper Palaeolithic. Some time between seventeen and fifteen millennia before the present, an inhabitant of what is now the Dordogne crushed haematite between stones, lit a pine torch, and rendered an aurochs on Lascaux limestone at approximately life size (Aujoulat, 2005). These visual arguments have continued to evolve. Twelfth-dynasty Egyptian painters at Beni Hasan committed a caravan of Semitic travellers, traditionally associated with the Joseph narrative, to a tomb wall around 1900 BCE (Kamrin, 2009); seventh-century Chinese astronomers committed over fourteen hundred stars to silk at Dunhuang (Bonnet-Bidaud et al., 2009); medieval cartographers compressed theology, geography and chronology onto the Hereford *mappa mundi* (Harvey, 1996). In 1786 William Playfair invented the bar chart in order to argue, graphically and in print, about Britain's trade balances (Playfair, 1786). John Snow's 1854 cholera map of Soho laid down the method of modern epidemiology (Johnson, 2006), and Charles Joseph Minard's 1869 flow map of Napoleon's 1812 Russian campaign (Friendly, 2002) is what Tufte (2001) has called, without serious contest, the densest argument ever made on paper (Figure 1.1a). When a human being intends a claim of any real quantitative weight, the first move is almost invariably to draw.

In contemporary practice, figures have not only evolved but have become significantly embedded in daily operations. Businesses read through dashboards of candlestick charts and cohort curves (Nison, 2001) that senior management consults in direct preference to prose; engineers coordinate the hundreds of thousands of components that constitute a ship, a hospital or an aircraft through hierarchies of blueprints read without ambiguity by people who may never speak to the designer (Ferguson, 1992); and even life-and-death decisions are made by reading a heartbeat off the grid of an electrocardiogram, or tissue contrast off a radiograph or perfusion map (Goldberger et al., 2017). Music (Treitler, 1984), satellite navigation (Kaplan and Hegarty, 2017), chemistry (Hoffmann, 1995) and network science (Newman, 2010) sustain the same diagrammatic primacy, and every competent empirical paper in the natural and social sciences eventually commits its argument to a figure (Figure 1.1b). **The figure has become the lingua franca of modern knowledge work**, and in many domains the argumentative weight it now carries is strictly greater than the weight carried by the accompanying prose.



(a)

The figure as the lingua franca of modern knowledge work



(b)

Figure 1.1: (a) A short genealogy of visual argument, from Lascaux and Egyptian wall painting through the medieval *mappa mundi*, Playfair’s 1786 bar chart and Snow’s 1854 cholera map to modern neural scaling curves. (b) The figure as lingua franca of modern knowledge work across business, music, medicine, engineering, navigation, chemistry, networks and the natural sciences. *Sources.* (a) Lascaux aurochs, Beni Hasan caravan, Dunhuang star chart, Hereford *mappa mundi*, Playfair 1786 bar chart, Snow 1854 cholera map and Minard 1869 flow map of Napoleon’s Russian campaign, all public domain (via Wikimedia Commons, the Metropolitan Museum of Art, the British Library, Hereford Cathedral, the Internet Archive and the Bibliothèque nationale de France respectively); scaling curves adapted from Kaplan et al. 2020. (b) A candlestick chart, score excerpt, ECG trace, blueprint, nautical chart, molecular diagram, network graph and scientific plot, from public-domain or openly-licensed repositories (Wikimedia Commons, NOAA, PhysioNet) or rendered by the author.

1.2 Can *Intelligentia Machinae* (Machine Intelligence) Comprehend Visuals Similarly?

Recent progress in artificial intelligence has been uncommonly rapid, and the current generation of large models has brought within plausible reach an older ambition of the field: a machine that can collaborate with a human being to achieve a shared task. A necessary condition of that collaboration is that the two parties reason on similar planes and navigate near-similar cognitive paradigms. Exact cognitive parity is not required; only a proximity close enough to sustain useful work in concert with a human collaborator. Since the human, as the previous section establishes, is a visual thinker, that proximity obliges the machine to comprehend the visual: to read, interpret and reason from the figure with fidelity comparable to its reading of text.

Systems at the current frontier, including Anthropic’s Claude Opus 4.6 and 4.7 (Anthropic, 2026a,b), OpenAI’s GPT-5 series (OpenAI, 2025a) and Google DeepMind’s Gemini 3 Pro (Google DeepMind, 2026a), are now deployed as infrastructure rather than laboratory curiosities. They are shipped into enterprise software through the major cloud providers, and they are measurably altering how code is written and how commercial operations are conducted. The salient question is therefore no longer whether such systems are capable, but whether that capability extends, in the way the human partner’s does, to the visual.

1.3 Motivation for Focusing on Scientific Research Figures

Research is among the first domains in which collaborative intelligence is being tested at scale: between late 2024 and early 2026 the three largest model developers each shipped autonomous research agents (Google, 2024; OpenAI, 2025c; Anthropic, 2025), and scientific deployments such as the Biomni biomedical platform now report research-cycle compressions of an order of magnitude or more (Anthropic, 2026d). Yet scientific knowledge is not monolingual; a substantial share of the record is produced in Chinese, German, Bulgarian and other languages of research, and the cost of excluding that share, in citation, uptake and participation, has been documented at length (Amano et al., 2023). Chart- and figure-understanding benchmarks, however, remain predominantly English: recent multilingual efforts such as PolyChartQA (Li et al., 2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English. The reach of a multimodal system, considered as a research collaborator, therefore scales with the breadth of languages over which its visual reading remains reliable, and the scientific figure, considered across languages, is the natural object at which to test whether the current frontier reads visually or merely appears to. What that reading must withstand, beyond correctness on the clean image, is illustrated in Figure 1.2.

The failure is not a miscount or a mislabelled axis; it is a confident substitution of a plausible category for evidence the model cannot actually see. Honest reading, not just accurate reading, is therefore the object of interest, and the research questions below are framed accordingly: they ask how faithfully today’s frontier models describe scientific figures across languages; how their reading survives visual degradation and adversarial context; and whether, when the evidence is not there, they acknowledge it rather than fill it in.

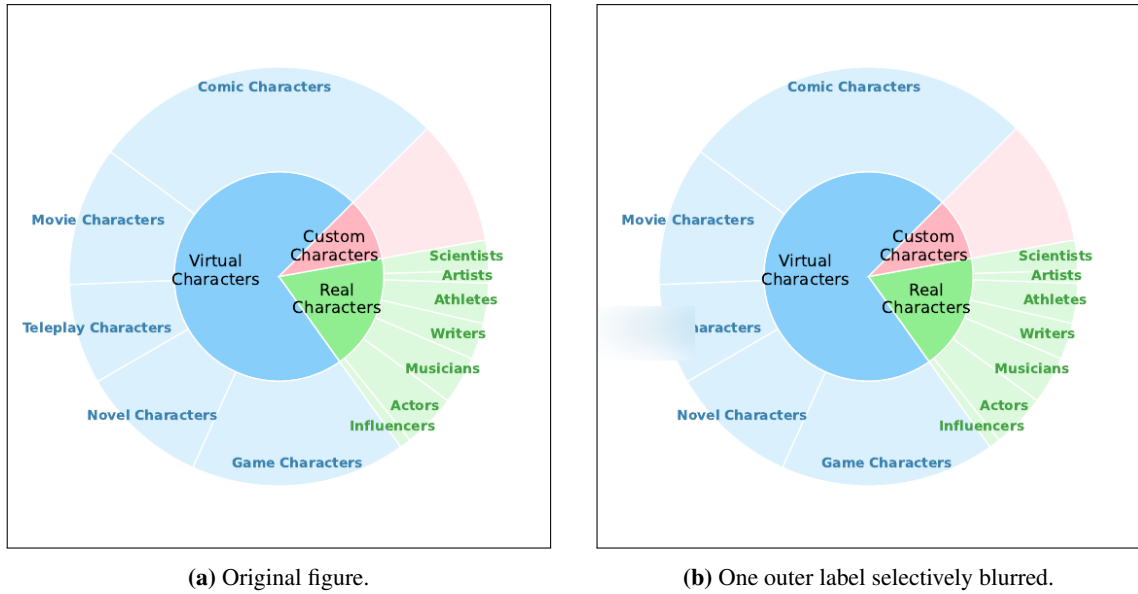


Figure 1.2: A sunburst taken from an arXiv paper in the SCIFIG-EVAL corpus. In (a), the outer-ring category between *Movie Characters* and *Novel Characters* is legibly *Teleplay Characters*; in (b), that single label has been blurred. Asked which category sits between the two, GPT-5.2 answers “*Anime Characters*,” with no admission that the relevant region is unreadable. Source image reproduced from arXiv:2505.23923v1 under its open-access licence; blurred variant generated by the SCIFIG-EVAL pipeline.

1.4 Research Questions

The central objective of this thesis is to characterise, across models, languages and adversarial conditions, how faithfully today’s frontier multimodal models actually read scientific figures, and to build an evaluation framework that surfaces the failure modes which conventional accuracy benchmarks obscure. Concretely, the investigation is organised around the following five research questions, each pairing a capacity with the conditions under which that capacity is tested:

RQ1 (Description accuracy across languages). To what extent do current vision–language models produce accurate and complete descriptions of scientific figures across typologically diverse languages?

RQ2 (Robustness to visual degradation). How does visual degradation, including noise, compression, aspect distortion and contextual embedding within paper pages, affect the accuracy (whether what is stated is correct), completeness (whether what is visible is reported) and clarity (whether what is reported is expressed unambiguously) of VLM-generated figure descriptions?

RQ3 (Comprehension beyond description). Beyond surface-level description, can VLMs demonstrate genuine comprehension of scientific visualisations, from answering quantitative questions to performing non-trivial inductive reasoning that requires distinguishing analytical insight from surface-level pattern matching?

RQ4 (Adversarial robustness). How robust are VLM outputs when subjected to adversarial manipulations, including misleading captions, contradictory premises and psychologically-informed sycophancy probes?

RQ5 (Calibrated epistemic honesty). When confronted with incomplete or unresolvable visual information, do VLMs exhibit calibrated epistemic honesty, acknowledging uncertainty rather than confabulating plausible but ungrounded responses?

1.5 Thesis Structure

The remainder of the thesis is organised as follows. Chapter 2 (Background) provides the technical and conceptual foundation: how frontier multimodal models convert figures into internal representations, how scientific figures carry argument rather than merely depict it, and the failure modes and evaluation traditions on which the remainder draws. Chapter 3 (Literature Survey) situates the work against chart understanding, multimodal hallucination, multilingual evaluation and MQM-style scoring. Chapter 4 (Design) sets out the SciFig corpus, the A–R–L framework and the five adversarial probe families. Chapter 5 (Implementation) describes the evaluation pipeline, the model and judge harness and the reproducibility controls. Chapter 6 (Evaluation) reports the main quantitative results across the 11 evaluated models and the four languages, together with ablations on judge ensembles and probe families. Chapter 7 (Discussion) interprets the findings, connects them to the A–R–L framework and acknowledges the limitations of the study. Chapter 8 (Conclusion) summarises the contributions, revisits the research questions and outlines directions for future work.

Chapter 2

Background

This chapter lays the technical and conceptual ground for the behavioural evaluation developed in Chapters 4–6. Section 2.1 describes how today’s frontier multimodal models turn an image into the representation on which every downstream capability rests. Section 2.2 treats the scientific figure as a carrier of argument rather than decoration, and distinguishes description from reasoning as cognitive acts. Section 2.3 surveys the three failure modes that shape this thesis — hallucination, sycophancy and miscalibration — and previews the Admittance–Resistance–Inductance framework that formalises them in Chapter 4. Section 2.4 introduces MQM-style open-ended evaluation and the LLM-as-judge methodology, and Section 2.5 closes on the complications of reading a figure whose text is not English.

2.1 How do today’s Large Language Models see?

To understand the visual capability of today’s Large Language Models, a useful starting point is to grasp how they see. Every family evaluated in this thesis shares the common skeleton sketched in Figure 2.1: a raster image is partitioned into patches by a vision encoder, the resulting embeddings are retuned by a connector into the token space of a language backbone, and the backbone consumes those projected vision tokens as a prefix alongside the text prompt (Liu et al., 2023; Alayrac et al., 2022; Li et al., 2023a). From the backbone’s point of view, the image is, by the time it arrives, a sequence of soft tokens drawn from the same vector space as text.

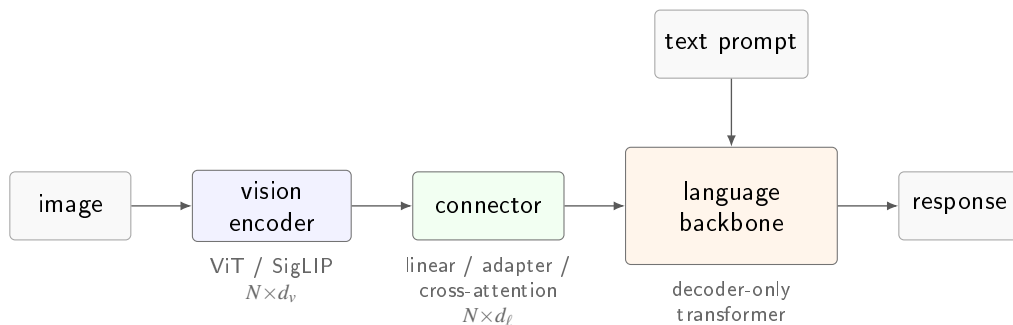


Figure 2.1: The canonical vision-language skeleton shared by every family evaluated in this thesis. A Vision Transformer (Dosovitskiy et al., 2021), typically trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, maps the image to N patch embeddings of width d_v ; a connector (linear projection, adapter, or cross-attention) retunes them to the backbone width d_l (Liu et al., 2023; Li et al., 2023a); the decoder-only language model consumes them as a prefix alongside the text prompt.

Within this skeleton, the seven families evaluated here diverge along four axes that turn out to

Table 2.1: Publicly documented vision-language architecture across the seven families evaluated in this thesis. “–” denotes a detail that is not disclosed in the primary source. The four open-weight families (top block) publish detailed technical reports; the three closed frontier families (bottom block) publish system cards that address capability and safety but leave the vision stack undisclosed. *Sources:* Gemma 3 (Gemma Team, Google DeepMind, 2025); Qwen3-VL (Qwen Team, Alibaba, 2025), with architectural continuity from Qwen2-VL (Bai et al., 2024); Llama 4 (Meta AI, 2025); Phi-4 multimodal (Microsoft Research, 2025); Claude Opus 4.x (Anthropic, 2026c); GPT-5 / 5.2 (OpenAI, 2025b); Gemini 3 Pro (Google DeepMind, 2026b).

Family	Vision encoder	Resolution / tiling	Image tokens	Fusion regime
Gemma 3	frozen SigLIP-400M	896×896 + Pan&Scan	256 (linear proj.)	adapter
Qwen3-VL	custom ViT + 2D-RoPE	dynamic, any aspect	variable (MRoPE)	native
Llama 4	MetaCLIP-derived ViT	–	– (early-fusion)	native early-fusion
Phi-4 mm	CLIP variant + LoRA	–	–	Mixture-of-LoRAs
Claude Opus 4.x	–	≤ 2576 px long edge	–	–
GPT-5 / 5.2	–	–	–	native (claimed)
Gemini 3 Pro	–	up to 900 images / prompt	–	native (sparse MoE)

matter empirically. The vision encoder is almost always a Vision Transformer (Dosovitskiy et al., 2021) trained with a CLIP (Radford et al., 2021) or SigLIP (Zhai et al., 2023) contrastive objective, except for Qwen3-VL (Qwen Team, Alibaba, 2025; Bai et al., 2024), which trains its own from scratch with 2D-RoPE to accept arbitrary resolution. The connector is either a linear projection (Liu et al., 2023), a cross-attention adapter with learnable queries (Alayrac et al., 2022; Li et al., 2023a), or LoRA modules grafted into the backbone itself (Hu et al., 2022; Microsoft Research, 2025). The fusion regime ranges from late adapter onto a frozen backbone (Gemma 3, LLaVA), through early-fusion on interleaved tokens (Meta AI, 2025; Chameleon Team, 2024), to ground-up native co-training (Qwen3-VL and the closed frontier families). The per-image token budget, finally, varies from a fixed 256 (Gemma Team, Google DeepMind, 2025) to aspect-preserving variable-length sequences (Bai et al., 2024). Table 2.1 collects what each family publicly discloses along these axes.

The table is as informative for what it omits as for what it contains. The four open-weight families each publish a technical report that documents the vision stack end to end; the three closed frontier families publish system cards (Anthropic, 2026c; OpenAI, 2025b; Google DeepMind, 2026b) oriented toward capability and safety, and leave the encoder, the resolution policy, the token budget and the connector architecture undisclosed. The asymmetry is structural: the models most widely deployed in real research settings are the models whose visual reading is architecturally least inspectable. An evaluation that required internal access would, at the frontier, have almost nothing to evaluate. This thesis therefore treats that constraint as a design requirement and, in Chapter 4, develops a three-axis behavioural framework — Admittance, Resistance and Inductance — that characterises a model’s visual reading from the outside.

2.2 The scientific figure as visual argument

Modern scientific knowledge is argued, not merely illustrated, on figures. Bertin (1983) codified the grammar of that argument through six visual variables — position, size, value, texture, colour and orientation — by which data acquire meaning on a page; Tufte (2001) formalised the accompanying principles of graphical integrity, namely proportionality between encoding and

value, and disambiguation between form and content; and Munzner (2014) provides the modern taxonomic apparatus that organises figures by the data relations they expose — rankings in bars, trends in lines, correlations in scatter, hierarchies in trees, flows in networks. The 1,005 figures in the SciFIG-EVAL corpus span this full taxonomy, and what unites them is not form but function: each asserts, with measured precision, a claim the surrounding prose expects the reader to verify.

For a vision-language model, the scientific figure poses difficulties that natural-image tasks do not. Its visual signal is almost entirely symbolic: axis labels, tick values, legend keys, error bars and annotations, rendered at small font sizes, often at low contrast, and embedded in dense visual grids that the encoder must read before the backbone can reason. Numeric tolerance compounds the difficulty, since misreading 0.42 as 0.47 on a bar chart is not the same kind of error as mistaking a cat for a dog. Huang et al. (2024) document exactly this pattern: frontier models that score well on natural-image benchmarks degrade sharply once the signal becomes largely textual and quantitative, a finding corroborated across CharXiv (Wang et al., 2024), ChartQAPro (Masry et al., 2025) and ChartMuseum (Tang et al., 2025).

Reading a figure is therefore two acts, not one. *Description* asks what is on the page: the value at a tick, the colour of a series, the label of a cluster. *Reasoning* asks what follows from it: whether a trend is monotonic, whether two series diverge, whether the stated conclusion is actually supported by the plotted data. The distinction is cognitive rather than linguistic: accuracy on description is compatible with baseless inference, and baseless inference is exactly the failure mode the behavioural probes of Chapter 4 are designed to elicit. The five research questions of Chapter 1 reflect this split — RQ1 and RQ2 probe description and its robustness under degradation, whereas RQ3 probes comprehension beyond the surface.

2.3 Failure modes of visual reading

Three failure modes recur across the vision-language literature and set the frame for the behavioural axes of this thesis.

Hallucination arises when a model produces content that is not supported by, or directly contradicts, the input image. Surveys of the phenomenon in language models (Ji et al., 2023) and its multimodal extension catalogue two regimes. The first is *object-level* hallucination, in which a model asserts the presence of an object that is not in the scene, as quantified by POPE (Li et al., 2023b) and HallusionBench (Guan et al., 2024). The second is *structural* hallucination, in which plausible but absent categories, labels or values are confabulated to fill gaps in the visual evidence; ChartHal (Xu et al., 2025) documents this failure mode directly on charts. Figure 1.2 in Chapter 1 illustrates the structural case: shown a sunburst with one outer label selectively blurred, GPT-5.2 supplies the confidently wrong label “Anime Characters” rather than admit that the relevant region is unreadable.

Sycophancy is the tendency to agree with a user’s implied position even when it is false. First formalised in the text setting by Sharma et al. (2023), it now has documented visual analogues: VLMs shift their description of a figure in response to leading questions, misleading captions or asserted prior beliefs, sometimes contradicting what they have just reported on the same image (Zhao et al., 2024). For a system deployed as a research collaborator, this is the failure mode with the largest downstream cost, because the stance a model should resist — the confident user who is

wrong — is the stance most often asserted in practice.

Miscalibration is a failure of self-knowledge: a mismatch between the model’s confidence and the evidence available to it. Kadavath et al. (2022) showed that language models can, under the right conditions, know roughly what they know; the visual setting has proven harder, with recent studies tying object hallucination directly to high epistemic uncertainty in early vision-encoder tokens and to disproportionate attention on a few “blind” patches. Calibration, on this view, is not a separate accuracy metric but the substrate on which honest abstention rests.

These three modes triangulate the behavioural framework developed in Chapter 4. *Admittance* names the readiness to acknowledge missing evidence (the absence of which is confabulation); *Resistance* names the ability to hold ground under misleading context (the absence of which is sycophancy); *Inductance* names the soundness of inference beyond surface pattern-matching (the absence of which is confident but baseless reasoning on top of a well-calibrated reading stack).

2.4 Open-ended evaluation and the LLM-as-judge

Accuracy-style benchmarks are the wrong instrument for the questions this thesis asks. A single-answer metric over a well-formed chart question cannot distinguish a model that reads carefully but abstains from one that guesses confidently; it cannot detect sycophantic drift across paraphrases; and it collapses to English in practice, because multilingual single-answer grading is brittle to translation. The open-ended description and reasoning tasks of Chapters 4–5 require a richer rubric.

Multidimensional Quality Metrics (MQM), introduced by Lommel et al. (2014) for machine translation, supplies one. MQM replaces a scalar score with an error typology (accuracy, fluency, terminology, style) in which each error is tagged with a severity weight (minor, major, critical), and aggregates the resulting annotations into a calibrated quality score. The same apparatus — typology plus severity — transfers directly to figure description, where the categories become visual fidelity, completeness and clarity, and severity separates a misread tick from a fabricated series.

At benchmark scale, human MQM annotation is infeasible, and the literature has converged on the *LLM-as-judge* paradigm (Zheng et al., 2023). The approach is fast, reproducible and, when carefully deployed, reaches human-level agreement; it is also subject to well-documented biases — self-preference, in which a judge rewards outputs from its own family (Wataoka et al., 2024); position bias in pairwise comparisons (Zheng et al., 2023); and length, verbosity and concreteness biases in scoring. The established mitigations — randomising positions, averaging over swaps, and aggregating across a heterogeneous judge ensemble — are adopted wholesale in Chapter 5. The pipeline used here draws from three judge families (Claude Opus 4.x, GPT-5 / 5.2 and Gemini 3 Pro) to blunt any single family’s preferences, and scores every candidate twice per rubric cell to cancel position bias.

2.5 Reading across languages

The scientific record is not monolingual. Chinese, German, Bulgarian and other languages host substantial shares of the primary literature, and the cost of excluding those shares — in citation, uptake and collaboration — has been documented at length (Amano et al., 2023). Chart-understanding benchmarks, by contrast, remain predominantly English: PolyChartQA (Li et al.,

2025), PM4Bench (Gao et al., 2025) and IndicVisionBench (Sharma et al., 2025) each report substantial cross-lingual degradation on tasks the same models solve comfortably in English.

Reading a non-English figure stresses a VLM at three layers. At the *pixel* layer, the encoder must resolve non-Latin script — Cyrillic superscripts, Chinese subscripts, German compounds that sit poorly in a tight axis label — through the same fixed patch grid it uses for English. At the *token* layer, the backbone’s tokeniser fragments non-English strings into finer subword units, attenuating any learned lexical prior. At the *prior* layer, the backbone’s world knowledge is itself skewed toward English-dominant pre-training, so that the completion a confident model gravitates toward is systematically more likely in English than in Bulgarian. Gao et al. (2025) show, on a parallel multilingual corpus, that higher-level figure reasoning is in practice bottlenecked by the VLM’s OCR across languages.

The three non-English languages chosen for SCIFIG-EVAL (Bulgarian, German, Chinese) were picked to cover this ladder: a Slavic Cyrillic language in Bulgarian, a high-resource Germanic Latin-script language in German, and a Sinitic logographic language in Chinese. Together they span script families, morphological complexity and pre-training density, and so isolate where in the stack each model family breaks.

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